



BSc Computer Science

School of Computer Science and Informatics, Cardiff University

CM3203 – One Semester Individual Project

**Exploring the Impact of Competition and Cooperation on Performance
and Engagement in VR Exergaming**

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Abstract

Acknowledgements

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1. Introduction

1.1. Context and Motivation

Exergaming, short for exertion gaming, is defined as technology-driven physical activities that require participants to be exerting themselves in one way or another in order to play a game. Exergaming was initially conceptualised in the late 1980s and gained significant popularity with the introduction of games like “Dance Dance Revolution” in the late 1990s. The introduction of the Nintendo Wii and Xbox Kinect in the early 2000s further pushed exergaming into the mainstream media, with these consoles and their games demonstrating the potential of exercise in gaming. The term “exergaming” entered the Collins English Dictionary in 2007, marking a point where the concept was officially recognised and popularised. Exergaming has continued to develop into not only video games, but also gaming technologies integrated with exercise equipment, mobile apps, and virtual reality. Virtual reality has become a compelling platform for exergaming development due to its unparalleled immersion and captivating nature. VR in exergames has shown to increase both enjoyment and motivation of physical exercise over both short (Farrow et al., 2018), and long periods of time (Michael et al., 2020). Virtual reality allows for a wider range of virtual activities compared to flatscreen exergaming, including sports, dance, and adventure games. Additionally, VR environments are promising treatment to mitigate functional decline and promote cognitive rehabilitation (Marotta et al., 2022).

1.2. Problem Statement

Physical inactivity has emerged as a critical global health crisis, affecting 31% of adults worldwide (1.8 billion people) as of 2022. This is a 5% increase since 2010, and it is predicted to reach 35% by 2030. Physical inactivity is a leading cause of chronic diseases like cardiovascular conditions, type-2 diabetes, some cancers, and dementia. It also causes muscle atrophy, which is particularly concerning for the 81% of UK office workers who spend 4-9 hours daily at their desks. Mental health is equally affected, with clear evidence showing that exercise prevents or at least reduces the severity of common disorders like depression and anxiety.

The global cost of chronic diseases is predicted to reach \$47 trillion by 2030. In workplaces physical inactivity contributes to decreased productivity, increased absence from work, and cognitive impairments that affect performance. The WHO has identified physical inactivity as the fourth leading risk factor for global mortality, contributing to 6% of deaths worldwide. Despite the growing popularity of VR exergaming as a tool to promote physical exercise, many VR fitness applications lack dynamic social interaction implementations that effectively sustain user motivation and engagement, while research has shown that incorporating competition and cooperation in exergames can both enhance user performance and effort (Michael et al., 2020), we have a limited understanding of how these social dynamics specifically impact user experience within VR cycling. Furthermore, more existing VR exergames are not tailored towards the user’s specific fitness level, resulting in an experience that may be either too difficult or too easy, reducing long-term engagement. Therefore, there is a critical need for further exploration on how competitive and cooperative elements in VR cycling applications influence the user’s performance, motivation, and engagement across a diverse range of users. Additionally, the integration of adaptive AI opponents and teammates that respond to the user’s effort could provide a scalable solution for tailoring difficulty and social interaction, making VR exergaming more accessible and enjoyable. Addressing these gaps is essential for developing VR fitness solutions that are immersive, personalised, and sustainable in promoting physical exercise.

1.3. Aims of the project

The aim of this project is to develop a VR cycling game with varied social interaction designs to investigate how cooperative and competitive gameplay mechanics influence user performance, engagement, and fitness outcomes. The project seeks to address the lack of accessible and engaging VR cycling platforms that cater to users with different motivational triggers and fitness levels. By integrating adaptive AI-driven scenarios and real-time feedback, the project aims to create an exergaming experience that is both engaging and inclusive for a wide range of users.

This project aims to develop three distinct cycling scenarios. These scenarios will take place in a prototype velodrome with minimal visual clutter to distract from the workout with the user completing the same physical workout routine through each scenario. (1) A baseline scenario involving the user pedalling around the velodrome on a solo bike ride, (2) a cooperative scenario involving the user pedalling around in a paceline with three AI teammates, (3) a competitive scenario involving the user pedalling around in a race competing against AI's with performance based on the users previous one or two scenario performances, with relative difficulty varying based on the users current heart rate.

This project aims to incorporate social interaction and gamification elements into VR exergaming through implementing features such as:

- Real-time performance tracking for the user to view if desired.
- AI teammates and opponents capable of dynamic behaviour adjustments based on user effort and exertion to maintain challenge and engagement.
- Post-race analytics to provide users with detailed performance feedback across all scenarios.

This project aims to evaluate user performance and engagement through a user study to assess performance metrics such as cadence, power output, heart rate, and time-to-completion across the different scenarios. Additionally, this project aims to gather qualitative feedback through surveys and questionnaires to evaluate user engagement, motivation, and experience in each scenario, as well as investigate possible correlations between user personality traits and preferences for competitive or cooperative gameplay modes.

Accessibility and usability are high priorities for this project. The user interfaces are designed to be intuitive, optimizing ease of use. The system also implements adaptive AI difficulty scaling to tailor challenge levels to users with varying levels of fitness. Additionally, it follows UI/UX best practices to ensure that users can navigate the system effectively, without unnecessary complexity or barriers.

1.4. Intended Beneficiaries

- **General fitness enthusiasts and individuals seeking motivational exercise tools** – For people looking for more engaging ways to exercise, especially those who struggle with traditional fitness routines due to lack of motivation or boredom, VR exergaming can be a great solution through its gamification of real-world exercises. Individuals of varying fitness levels, from beginners to advanced athletes, who can benefit from adaptive AI that tailor's difficulty to their abilities.
- **Individuals interested in home-based or VR-based fitness solutions** – Home exercisers, especially those with limited access to gyms or outdoor activities, who are seeking immersive and interactive fitness experiences. People motivated by interactive environments and social gaming elements to maintain regular exercise habits.
- **Health and rehabilitation professionals** – Physical therapists and rehabilitation specialists looking for innovative ways to encourage cardiovascular exercise and rehabilitation training through VR. Could be adapted for patients needing low impact yet engaging cardio sessions in controlled environments.

- **Researchers and developers in VR, exergaming, and health technology** – Academic researchers exploring the psychological and physiological impacts of social dynamics in exergames, such as cooperation and competition. Game developers and VR designers focusing on serious games for health, who can use insights from this project to create more adaptive and motivational exergaming experiences.
- **Organisations focused on public health and fitness promotion** – Public health bodies, fitness organisations, and wellness programs aiming to promote active lifestyles using innovative technologies like VR. These organisations could leverage motivational insights from this project to design campaigns or interventions encouraging physical activity through gamified VR experiences.
- **The VR and gaming industry** – VR developers seeking to expand into the fitness and wellness market with products that combine entertainment with physical benefits. Companies working on hardware that could integrate social and adaptive features into their platforms.

1.5. Project Scope and Limitations

This paper focuses on the development and evaluation of a VR cycling exergame designed to explore the impact of social dynamics on user engagement, motivation, and performance. The study investigates how adaptive AI teammates and opponents can enhance the VR exergaming experience and maintain player motivation through real-time difficulty adjustments based on effort and exertion levels. The scope of this project includes:

- **VR Cycling Simulation** – Development of a virtual velodrome cycling environment with three distinct gameplay modes: Baseline, Cooperative, Competitive. With clear UX design allowing users to navigate to their desired experience.
- **Social Interaction Elements** – Implementation of AI-driven teammates and opponents that adapt dynamically to the user's performance.
- **Engagement & Performance Assessment** – Evaluation of user motivation, effort, and fitness metrics (speed, cadence, heartrate, power output) through quantitative and qualitative methods.
- **Gamification & UI/UX** – Design of intuitive interfaces, real-time performance tracking, and post session analytics to enhance user experience.
- **Adaptive Difficulty Scaling** – Integration of AI-based adjustments to tailor challenge levels according to user fitness levels and exertion.

Project Limitations:

- **Limited Participant Pool** – The study involved in this project does not account for all demographics, such as older adults or individuals with disabilities.
- **Short-term Study Duration** – The research evaluates immediate engagement and performance rather than long-term fitness adherence.
- **Single-Modality Exergame** – The focus is solely on VR cycling, without exploring other exercise forms such as running or strength training.
- **Absence of Multiplayer Mode** – While the study explores AI-driven social interaction, real-time multiplayer experiences may be more effective, however are beyond the scope of this project.
- **Absence of Environmental Modifications** – This project revolves solely around a velodrome map prototype and does not take into account how changes in environment and surroundings impacts engagement, motivation, and performance.

1.6. Research Questions

- How do competitive and cooperative dynamics impact user performance in VR Exergaming?

- Does competition or cooperation lead to higher engagement and motivation compared to solo cycling?
- How do different personality types respond to competitive vs cooperative VR exergaming scenarios?
- Can AI-driven real-time adjustments improve user experience and maintain challenge levels across different fitness levels?
- What role do social interaction mechanics play in sustaining long-term engagement with VR-based fitness activities?

2. Background

2.1. VR exergaming landscape

2.1.1. Competition and Cooperation in Exergaming

Exergaming is a field which has gained traction in research over the last 10-15 years. Studies have suggested that exergames have the potential to motivate inactive individuals to exercise (Warburton et al., 2007), but that they can also struggle with prolonged adherence similar to conventional exercise (Mestre, Dagonneau & Mercier, 2011). The focus on this project, however, is to see how changes in social dynamics in cooperative and competitive environments impact exercise adherence, enjoyment, motivation, and performance, and to additionally see if results correlate with particular personality types favouring different dynamics.

Previous studies show that competitive and cooperative factors in exergames influence user motivation. Peng & Crouse (2013) used an exergame possessing three different scenarios, (1) A single player mode where users played against their self pretest score; (2) a cooperative mode with another player in the same physical space; (3) a mode with another player in parallel competition in separate physical spaces. In this study they found (3) parallel competition to be the optimal mode, resulting in high enjoyment and future play motivation and high physical intensity. The study also found that cooperation was more enjoyable and motivating than solitary play, but the singleplayer mode led to greater levels of physical exertion.

Competition can have a strong impact on intrinsic motivation (Song et al., 2009), a study that explored both highly and lowly competitive individuals in both highly and lowly competitive conditions found that intrinsic motivation could be increased for highly competitive individuals when presented with competitive exergaming settings, however that for less competitive individuals being placed in a competitive environment suffered detrimental effects to their exercise experience, reducing their enjoyment and their likelihood of engaging in further sessions.

Cooperative dynamics have also exhibited benefits in the field of exergaming. A study on Adolescent exergame play for weight loss and psychosocial improvement (Staiano, Abraham & Calvert (2013) had participants from a public high school randomly assigned to different modes (competitive, cooperative, control) within the Nintendo Wii Active game for 30-60 minutes. Results showed that players in the cooperative exergame lost significantly more weight than the control group which didn't lose any weight. The competitive players did not differ significantly from the other conditions. It was also found that cooperative players significantly increased in self-efficacy compared to the control group and both competitive and cooperative players significantly increased in peer support. This concluding that exergames, particularly in cooperative settings can be an effective tool for weight loss among youth.

Recent cross-sectional studies further examined the effects of competition and cooperation on exercise performance in immersive virtual reality cycle exergames (shaw et al., 2016). In one study, competing against a ghost of the user's previous session resulted in a greater distance travelled by the user compared to cycling with a virtual trainer present. Additionally, the ghost mode was rated as more enjoyable than both solo cycling and cycling with a virtual trainer. A follow-up study revealed that a competitive virtual trainer system elicited greater distance travelled and caloric expenditure compared to a cooperative virtual trainer system and was rated as more motivating. Interestingly, in both studies, self-reported enjoyment and motivation did not significantly correlate with actual performance metrics (distance travelled, and calories burned). These findings reinforce the effectiveness of competitive elements in

exergaming as tools to elicit higher levels of exercise, demonstrating that competitors can successfully substitute for human opponents in driving improved performance.

2.1.2. Interactive Feedforward in Exergaming

Interactive feedforward is an interactive adaptation of the psychological feedforward training method where rapid improvements in performance are achieved by creating self-models showing previously unachieved performance levels (Barathi et al., 2020). It can be used to enhance performance, understanding, and engagement by providing users with previous of potential outcomes and guiding them towards better results.

Unlike traditional feedback, which focuses on past performance, feedforward focuses on future possibilities and potential outcomes. Its about anticipating what might happen and guiding users toward desired results. Interactive feedforward takes this concept further by incorporating real-time interaction and feedback within the system. This allows for dynamic adjustments and improvements as the user progresses.

A HIIT (High Intensity Interval Training) 4-week longitudinal study on exergaming with the use of interactive feedforward had user's complete workouts with no ghost condition, and with a ghost condition where logged performances were put into a ghost non-player character with an increased performance of 5% to visualise the future potential performance of the user. During this study participants who engaged in the non-competitive scenario made an average performance gain of 8%, compared to the participants who engaged in the competitive scenario using interactive feedforward made an average performance gain of 16%.

Additionally, participants in the competitive (Multi-Ghost) group found that they were more intrinsically motivated, had a greater perceived competence, and put in a greater amount of effort, compared to the solo cycling group (Non-Competitive) (Michael et al., 2020). Overall, this study proved that interactive feedforward can be an effective method of improving player performance over time.

A study in 2018 examined the application of interactive feedforward in high-intensity VR exergaming, addressing the gap in effective protocols for high intensity exercise in virtual environments. Researchers tested the interactive feedforward training technique by having participants cycle against either models of their own selves' previous performances, or virtual competitors, with varying levels of resistance and awareness. The results demonstrated that interactive feedforward successfully enhanced participant performance while preserving intrinsic motivation, even when participants were aware of the interventions. Notably, competing against modified self-models proved more effective than competing against virtual competitors. These findings suggest that interactive feedforward offers a promising approach for designing high-intensity VR exergames that can simultaneously improve physical performance and maintain user engagement (Barathi et al., 2018).

2.2. Effects of a lack of exercise

2.2.1. Physical Health Consequences

Physical inactivity is a primary cause of most chronic diseases (Booth, Roberts and Laye, 2012). In 2022 31% of adults worldwide (1.8 billion people), did not meet the recommended levels of physical activity (World Health Organization, 2024). This is an increase of 5 percent points between 2010 and 2022. If this trend continues, the proportion of adults not meeting recommended levels of physical activity is projected to rise to 35% by 2030 (World Health Organization, 2024). Physical inactivity puts adults at greater risk of cardiovascular diseases such as heart attacks and strokes, type 2 diabetes, dementia and cancers such as breast and colon. Additionally, physical inactivity can cause physiologic muscle atrophy as people's muscles aren't being used enough. This is particularly a problem for those who have seated jobs (Medline Plus, 2019), and considering it is claimed 81% of UK office workers spend between four and nine hours each day sitting at their desk (Bean, 2018), this is a considerable concern for a large amount of the population.

2.2.2. Mental Health Impacts

Mental illness has become a major health problem worldwide, particularly during COVID-19 it was estimated that approximately 12% of the global population had various mental illnesses, which was associated with a significant burden on healthcare systems like the NHS translating into approximately 5% of disability-adjusted life years lost (Schuch and Vancampfort, 2021). This research found compelling evidence that physical activity and exercise can prevent common mental disorders, such as depression and anxiety disorders, and have multiple beneficial factors on the physical and mental health of people who already have mental health disorders.

2.2.3. Societal and Economic Costs

According to the World Health Organization (WHO), 63% of worldwide deaths in 2008 were caused by major preventable diseases (GLOBAL ACTION PLAN FOR THE PREVENTION AND CONTROL OF NONCOMMUNICABLE DISEASES, 2013). In 2011, the cost of chronic disease worldwide was estimated to reach \$47 trillion by 2030 (Methodological Appendix: The Global Economic Burden of Non-Communicable Diseases, 2011). Physical inactivity at work leads to decreased productivity and increased absenteeism due to various health issues and cognitive impairments. Reduced physical activity can cause fatigue, musculoskeletal problems, and mental health issues, all of which contribute to time off work. Additionally, a lack of physical activity can impair cognitive function, making it harder for employees to focus and concentrate, impacting their productivity while at work. In terms of presenteeism, individuals who are not physically active lose a larger amount of working time (between 2,6 and 3,71 days per year) compared to an individual who is physically active (Hafner et al., 2019).

2.2.4. The Sedentary Lifestyle Trend

The World Health Organisation (WHO) identified physical inactivity as the fourth leading risk factor for global mortality. It contributes to 6% of global deaths. Following after high blood pressure (13%), tobacco use (9%), and high blood glucose (6%) (Nih.gov, 2010). VR exergames provide a unique opportunity to revolutionise traditional exercise routines in ways reality simply cannot aim to tackle the global issue of sedentary lifestyles. By merging immersive and motivating gaming experiences with pre-existing workouts, we can both increase enjoyment and motivation of exercise, as well as the convenience and accessibility of indulging in it.

2.3. Gamification and social interaction mechanics in VR fitness

In the Cambridge dictionary gamification is defined as “the practice of making activities more like games in order to make them more interesting or enjoyable” (Cambridge Dictionary, 2014). Gamification elements can include features such as points, challenges, levels, or badges in the contexts of exercise to increase motivation and enjoyment. Commercial VR exergames like Zwift, FitXR and Supernatural) use gamification heavily, but gamification isn't the only path to user engagement, this project focuses adopts a minimalist approach, focusing instead on performance feedback and social dynamics to increase engagement. This allows users to concentrate on exertion without visual clutter distracting them from the workout.

Rather than focusing on plain gamification, this system includes a heads-up display (HUD) for the user showing key workout metrics (speed, cadence, power, heartrate, time), which supports user engagement through self-awareness. This minimalist design aligns with the needs of users who may prefer data driven motivation rather than playful interaction. (Shaw et al., 2016) found that participants in VR cycling scenarios with cooperative and competitive social dynamics expended significantly more effort and reported higher engagement compared to those cycling alone or with a ghost. This supports the idea that

social presence and dynamics can act as standalone motivators in exergames, aligning with this project's focus on AI-driven social mechanics over traditional gamified elements. Similarly, (Michael et al., 2020) demonstrated that a multighost competition model in VR cycling led to a 16% increase in performance over four weeks. This reinforces the idea that gamification is not required to drive improvement, rather, engaging challenges that evolve with the user, such as AI-based competition, can yield meaningful fitness outcomes. According to Deci and Ryans Self-Determination Theory (Ryan and Deci, 2000), intrinsic motivation arises when individuals experience competence, autonomy, and relatedness. The heads-up display (HUD) used in this system directly supports competence through real-time feedback, while the cooperative and competitive modes encourage relatedness, creating a socially meaningful workout experience even with the absence of traditional gamified features.

While (Cugelman et al., 2013) outlines game mechanics such as rewards and progress tracking as key tools in behaviour change, the paper also emphasises the importance of timely feedback and goal-related metrics, both of which are central to the program developed in this project. By presenting performance stats through a heads-up display, the user remains engaged through self-monitoring, fulfilling many of the same behavioural incentives as gamification.

2.4. Existing gaps in VR exergaming systems and preexisting solutions

Most VR fitness apps offer a static challenge and rarely offer intelligent adaptability for different fitness levels. This project addresses this gap by using interactive feedforward to proactively adjust AI behaviour based on historical user data which tailors the workout challenge to push the user further than they did before.

Previous studies (Shaw et al., 2016) had users speed directly correlate to how fast the users cadence was, i.e. the faster you pedal the faster you go. This opens up research to flaws such as inconsistent pacing strategies across sessions where less physically active participants better learn how to prolong their workout simply by having more practice on the system. This project resolves this problem with the use of ERG mode, a feature on many smart bike trainers that automatically adjusts the resistance to help you maintain a specific power output rather than having to manually control resistance. Using ERG mode, this project was able to eliminate the variable of pacing through ensuring that the users were outputting the wattage the exercise bike was asking of them. Additionally, having participants produce a set wattage each second of the workout allows AI to better tailor the challenge to the participant as the data is smoother and more reliable when it comes to mimicking previous performances. Past research has shown competition, and cooperation can boost motivation (shaw et al., 2016, Michael et al., 2020), but few systems intentionally compare these dynamics in a controlled way. Additionally, there is little analysis of user preference or personality types influencing effectiveness. This solution directly compares scenarios (1) baseline, (2) cooperative, (3) competitive, and collects both quantitative performance feedback, and qualitative form feedback after all three. From this, this project aims to test which type of social interaction is most effective and investigate if different personalities thrive in different modes.

Many fitness games assume a general audience and don't scale well from complete beginners to high performance athletes. Additionally, particularly in the commercial industry, exergames often lack inclusive design for different motivational triggers or ability levels. This project combats that using the baseline condition to calibrate difficulty, as well as emphasises accessibility in UI and game pacing.

2.5. Justification for the study based on the literature and gaps identified

The literature highlights several opportunities for further innovation within the VR exergaming space. While prior research has shown that social dynamics such as competition

and cooperation enhance motivation, few systems have directly compared these interaction styles in a controlled environment, nor accounted for how personality traits influence their effectiveness. Furthermore, many existing exergames fail to dynamically adapt to varying fitness levels, resulting in static challenges that may demotivate users over time or fail to meet their needs.

This project addresses these gaps by integrating adaptive AI via interactive feedforward, utilising ERG mode to control pacing consistency, and calibrating challenge levels using initial baseline testing. By focusing on inclusive, data-driven design and comparing distinct social interaction modes, the system seeks to provide more meaningful engagement, better performance outcomes, and deeper insight into how different users respond to varied exergaming experiences.

3. Methodology / Approach / Specification, Design, and implementation

3.1. System specification and Functional Design

3.1.1. Goals for the VR exergame

The VR cycling exergame system is required to facilitate the investigation of how different social dynamics affect user performance, engagement, and motivation during exercise.

Specifically, the system must:

- Enable controlled comparison of three distinct cycling scenarios (baseline, cooperative, competitive) while maintaining consistent workout parameters.
- Capture comprehensive performance data across all scenarios to enable comparative analysis.
- Adapt dynamically to users' fitness levels through AI-driven opponents/teammate behaviour.
- Present relevant information to users via an intuitive interface that minimises distraction.
- Support post-session analysis of performance metrics and user experience.
- Accommodate users of varying fitness levels and exercise experience.

These requirements are derived directly from the project aims outlined in section 1.3, which seek to address the identified gaps in current VR exergaming solutions regarding adaptive AI and varied social interaction designs.

3.1.2. Description of the three scenarios

The system implements three distinct cycling scenarios within a consistent virtual environment. Each scenario represents a different social dynamic while requiring the same profile of physical exertion from the user.

Baseline Scenario

This scenario serves as a control condition and consists of:

- A solitary cycling experience in the velodrome environment
- No AI entities present, only performance tracking
- User focused entirely on their own performance metrics
- Data from this session used to calibrate subsequent scenarios

Cooperative Scenario

This scenario implements a cooperative team dynamic through:

- Integration with three AI teammates forming a paceline formation
- AI behaviour programmed to maintain tight formation with the player
- AI adapted to ensure pace is consistent with player, as well as player pace consistent with AI.

Competitive Scenario

This scenario creates a competitive environment through:

- Racing against AI opponents calibrated to the users' baseline performance and or cooperative performance (historical data)
- Dynamic difficulty adjustment based on heart rate monitoring
- Performance-based positioning
- Use for interactive feedforward to push player beyond their previous best

3.1.3. Consistent environmental settings.

To ensure valid comparison between sessions, the following elements remain consistent:

Virtual Environment

Each scenario uses the same prototype velodrome environment (Sketchup.com, 2017) that was modified to flat colours for purposes of texture simplicity, as well as to reduce cognitive

load and maintain focus on exercise. In this environment lighting conditions and visual clarity remained consistent across all three scenarios.

Exercise Parameters

Each participant completed the same “Max Ramp Test” workout from Cardiff University’s VR Lab Wattbike Atom 2 exercise bike, giving them identical power output targets across all scenarios. ERG mode was used on the Wattbike to ensure consistent resistance regardless to participant effort.

User Interface Layout

Each scenario used the same heads-up display (HUD) to deliver performance metrics to the participants in real-time, uniform font sizes and notification systems, and standard positioning of critical information in the user’s field of view.

3.1.4. Performance tracking elements.

The system captures and visualises several key performance metrics.

- **A Real-time Metrics Display (HUD)** displaying current power (watts), current cadence (revolutions per minute), current heart rate (beats per minute), current speed (miles per hour), and elapsed time (seconds).
- **Post-Session Analytics** through performance graphs to show trends over time, as well as comparison tools developed in python for cross scenario analysis. Additionally, statistical summaries are generated at the bottom of performance log spreadsheets to draw quick conclusions from workout averages and other metrics.

3.2. Methodology, Requirements and Tools

3.2.1. Software Development Methodology

This project took an iterative development approach with progressive refinement. Core functionalities were initially developed sequentially to establish a barebones working prototype. Once this state was achieved, each feature was periodically revisited for refinement based on peer feedback. This maintained flexibility of development making adapting to changing requirements based on project feasibility and technical roadblocks much easier.

3.2.2. User Requirements

Functional Requirements

Exercise Interaction

- The system must accurately track cycling cadence, power output, and heart rate in real time.
- Users must be able to control their virtual cycling experience through physical pedalling.
- The system must interface reliably with standard smart bike trainers.
- The system should record all exercise metrics for post session analysis

VR Experience

- Users must be able to view the virtual velodrome environment with proper depth perception and scale.
- The system must maintain a consistent framerate to prevent motion sickness.
- Users must be able to look around naturally within the VR environment.
- The system should minimise visual elements that could cause discomfort in VR.

Scenario Management

- Users must be able to easily select between the three cycling scenarios.
- The system must provide clear instructions to the user for how to run each scenario.
- Users must be able to start and end sessions as needed.
- The system should provide appropriate transitions between game states (e.g. menu to workout).

Performance Feedback

- Users must be able to view critical performance metrics (speed, cadence, power, heart rate) in real time.
- Users must be able to access detailed performance analytics after each session.
- The system should allow comparison of performance across different scenarios.

AI Interaction

- In the cooperative scenario, AI teammates must respond appropriately to the user's pace and position.
- In the competitive scenario, AI opponents must provide an appropriate challenge based on the user's baseline performance.
- The system must adjust AI difficulty based on the user's heartrate and exertion level.
- The system should provide realistic AI behaviour that enhances immersion.

Non-Functional Requirements

Usability

- The system interface must be understandable without prior training.
- Users must be able to navigate all system functions while wearing a VR headset.
- The system should minimise physical setup time to under 5 minutes.
- Text and UI elements must be legible in the VR environment.

Performance

- The system must maintain consistent performance throughout the workout.
- Input lag between physical pedalling and virtual movement must not exceed the minimum time that Bluetooth data is received.
- The system should efficiently process and store performance data without noticeably impacting VR performance.

Reliability

- The system must function consistently across multiple sessions without requiring recalibration.
- The system must safely handle potential hardware disconnections during use.
- The system should provide appropriate error messages for any system failures.

Data Management

- The system must securely store all user performance data.
- Users must be able to access their historical performance data.
- The system should export data in standard formats for analysis.
- Personal identifying information must be properly anonymised for research purposes.

3.2.3. Development environment

The project setup phase of the project involved becoming familiar with the facilities available, that being the VR Lab in the School of Computer Science and Informatics. This lab possessed the necessary hardware to complete this project, including Polar H10 Heartrate Monitors, Meta Quest 2's, a Wattbike Atom 2 Exercise Bike, and a powerful desktop computer.

For the simplicity of the user, this project is entirely controller free utilising Unity's hand tracking integration. This prevents hassle with users using controllers whilst cycling, as their hands should be as comfortable on the exercise bike handlebars as possible.

Despite Meta Quest 2's offering standalone android deployment, for the sake of simplicity in prototyping this project is a PCVR build.

Unity Packages Used:

- AI Navigation
- TextMeshPro
- UnityUI

- XR Core Utilities
- XR Hands
- XR Interaction Toolkit
- OpenXR Plugin
- Oculus XR Plugin
- XR Plugin Management

3.3. Development

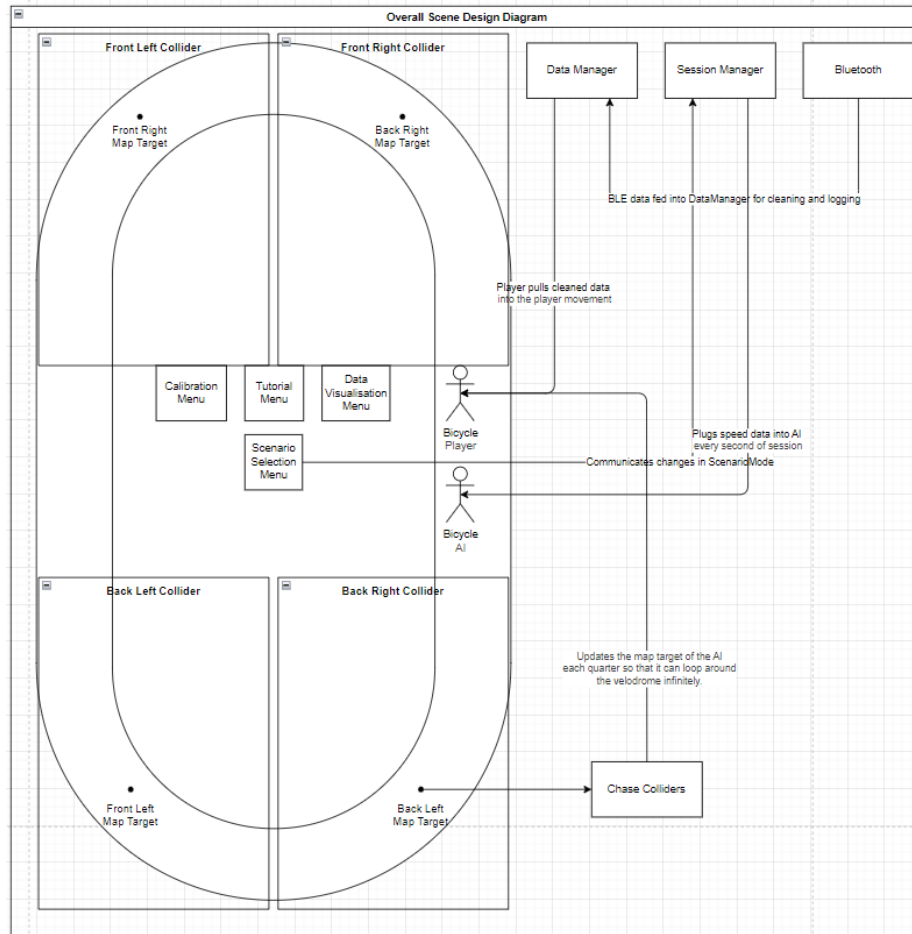


Figure 1 - Simplified Scene Layout Diagram

3.3.1. User Interface and User Experience Menus

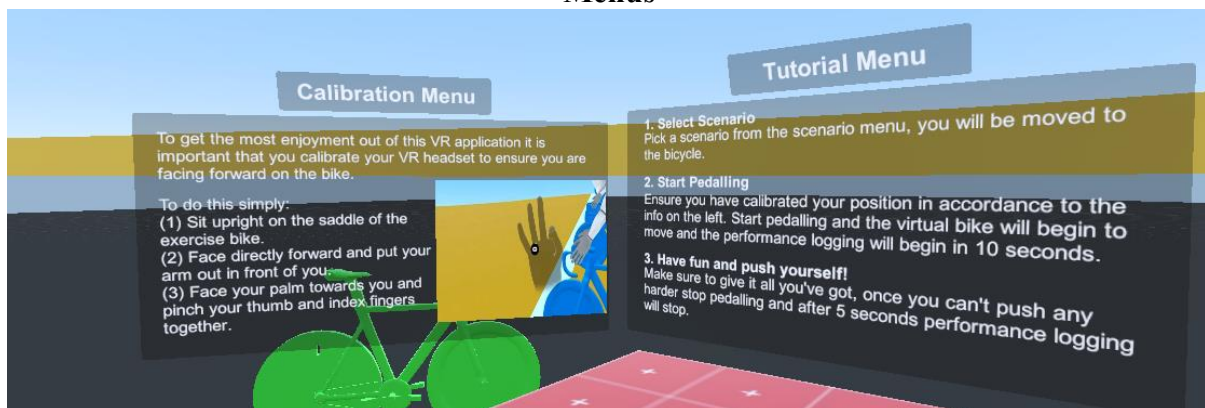
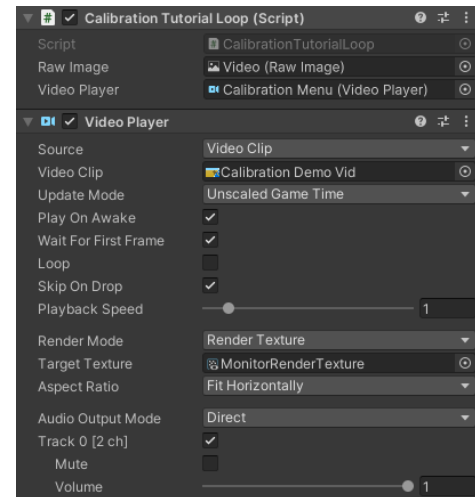


Figure 2 - Calibration & Tutorial Menu

The main menus the user uses are the calibration and the tutorial menu. The calibration menu is used to explain to users how to calibrate their headset before the workout to ensure that they are positioned facing forward in the unity scene and their hands land on the handlebars of the 3D bicycle model to improve immersion. The menu also uses a video recorded of what the calibration procedure looks like for the Meta Quest 2 using Unity's Video Player component as well as the script CalibrationTutorialLoop.cs I wrote. This script is simply responsible for assigning the VideoPlayers target to the Unity RawImage in the menu.



The tutorial menu is a static menu presented directly in front of the user which explains the game loop in a digestible way, explaining how the experience works so that they can get the most out of their session.



Figure 3 - Scenario Selection Menu

The scenario selection menu is the centre hub of navigation for the user as it determines which scenario they will enter for their workout. Each button uses Unity's built-in XR Poke components and once clicked communicate with the SessionManager to set the scenario mode of the session. More on that in 3.3.2. Singleton Managers.

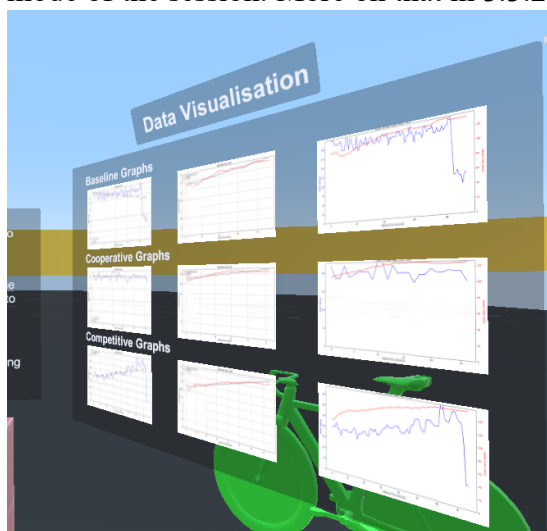


Figure 4: Data Visualisation Menu

The data visualisation menu is responsible for presenting the users performance across the various scenarios. For each scenario a (1) Power graph, (2) Heart rate graph, (3) Combined graph, is generated. These graphs are also saved as .png images at the user's devices persistent data. For this feature to work the project contains two data visualisation scripts, DataVis.py and UnityGraphLoader.cs. The data visualisation module of this project begins within the UnityGraphLoader.cs. Within this script is a public method called DataVisualisationScript() which takes the script name, the csv file names, and the output directory to save images in. It can be called through a UI button on the Developer Flatscreen UI implemented for convenience

during prototyping and a failsafe for if Bluetooth devices disconnect mid workout, or automatically after a workout is finished.

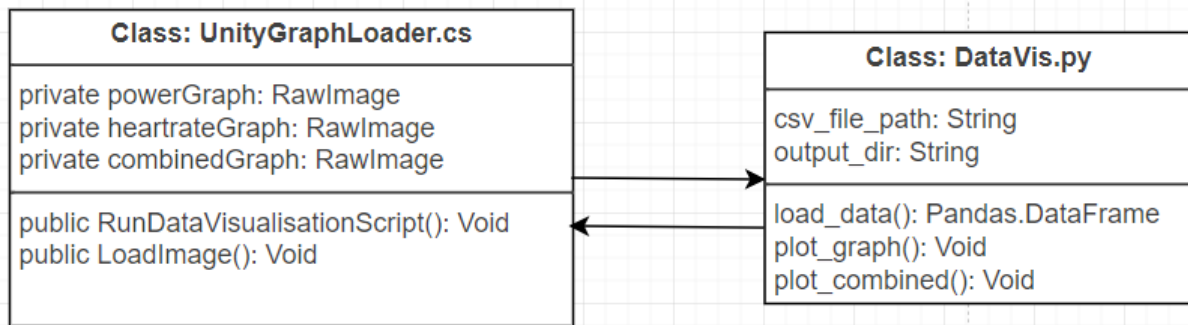


Figure 5 - Interaction between Data Vis Classes

```

public void RunDataVisualisationScript(string scriptName, string csvFileName, string outputImageFolder)
{
    // Define all the necessary paths
    string scriptPath = Path.Combine(Application.streamingAssetsPath, scriptName);
    string csvPath = Path.Combine(Application.streamingAssetsPath, csvFileName);
    string destScriptPath = Path.Combine(Application.persistentDataPath, scriptName);
    string destCsvPath = Path.Combine(Application.persistentDataPath, csvFileName);
    string imagePath = Path.Combine(Application.persistentDataPath, outputImageFolder);

    // Copy Python script to persistentDataPath if needed
    if (!File.Exists(destScriptPath))
    {
        File.Copy(scriptPath, destScriptPath);
    }

    // Copy CSV file to persistentDataPath if needed
    if (File.Exists(csvPath) && !File.Exists(destCsvPath))
    {
        File.Copy(csvPath, destCsvPath);
    }

    // Create process to run Python script
    ProcessStartInfo psi = new ProcessStartInfo();
    psi.FileName = "python"; // Uses Python from PATH
    psi.Arguments = $"\"{destScriptPath}\" \"{destCsvPath}\" \"{imagePath}\""; // Pass both CSV and image paths
    psi.RedirectStandardOutput = true;
    psi.RedirectStandardError = true;
    psi.UseShellExecute = false;
    psi.CreateNoWindow = true;

    UnityEngine.Debug.Log($"Command: python \"{destScriptPath}\" \"{destCsvPath}\" \"{imagePath}\"");

    // Execute Python process
    using (Process process = Process.Start(psi))
    {
        string output = process.StandardOutput.ReadToEnd();
        string error = process.StandardError.ReadToEnd();

        process.WaitForExit();

        if (!string.IsNullOrEmpty(output))
            UnityEngine.Debug.Log($"Python output: {output}");

        if (!string.IsNullOrEmpty(error))
            UnityEngine.Debug.LogError($"Python error: {error}");

        UnityEngine.Debug.Log("Python script execution completed.");
    }
}

```

Figure 6 - RunDataVisualisationScript(...) - UnityGraphLoader.cs

This script works by establishing the paths of operations on the user's local system by using Unity's Application.persistentDataPath variable ensuring that the script works on different systems and copies the script and spreadsheets to the persistentDataPath if necessary to ensure that it works at runtime. Once that is done the script creates a new Process from the System.Diagnostics namespace and executes the necessary command running the python script DataVis.py and passing through the desired parameters, giving debug logs on the process status as it goes along.

```

public void LoadImage(RawImage img, string imagePath)
{
    if (File.Exists(imagePath))
    {
        byte[] fileData = File.ReadAllBytes(imagePath);
        Texture2D texture = new Texture2D(2, 2);
        texture.LoadImage(fileData);
        img.texture = texture;
        UnityEngine.Debug.Log($"Image successfully loaded from: {imagePath}");
    }
    else
    {
        UnityEngine.Debug.LogError($"Image not found at: {imagePath}");
    }
}

```

Figure 7 - LoadImage(...) - UnityGraphLoader.cs

The DataVis.py script is responsible for the plot processing of the graphs that are used in the UnityGraphLoader.cs script. The command within the Unity script is responsible for running this python file at runtime as Unity does not natively support it.

```

# Main execution
if __name__ == "__main__":
    df = load_data(csv_file_path)
    os.makedirs(output_dir, exist_ok=True)

    plot_graph(df["ElapsedSeconds"],
               df["Power"], "Elapsed Time (seconds)", "Power (Watts)", "Power Over Time", "blue",
               os.path.join(output_dir, "power_graph.png"), 0, df["Power"].max())
    plot_graph(df["ElapsedSeconds"],
               df["HeartRate"], "Elapsed Time (seconds)", "Heart Rate (BPM)", "Heart Rate Over Time", "red",
               os.path.join(output_dir, "heartrate_graph.png"), 0, None)
    plot_combined(df, os.path.join(output_dir, "combined_graph.png"))

    print("Script completed successfully")

```

Figure 8 - Main Execution - DataVis.py

The script comprises of three stages, (1) loading data from the CSV files generated after the workouts, (2) Making the directory for the graphs to be stored in ensuring that they will be picked up by the Unity script, (3) plotting the graphs. Once each workout is complete the script will generate three graphs, making nine in total to be presented to the participant once they have completed the full study.

Heads Up Display (HUD)

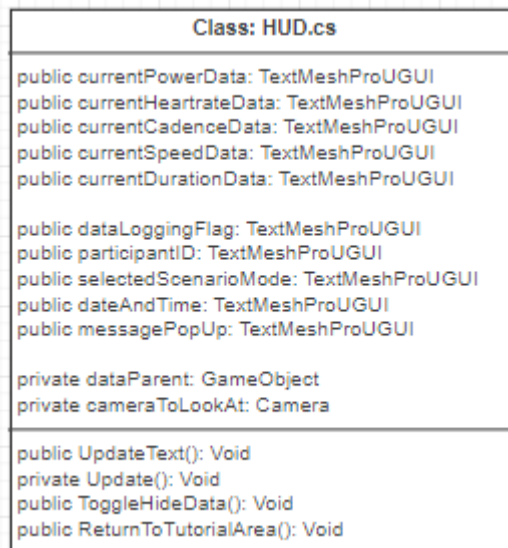


Figure 9 - HUD Class Diagram

The heads-up display is the sole user interface for the player whilst they participate in their workout. They are presented with critical information in real-time as well as feedback from the data logging system and session information. Within the Unity project the HUD is ultimately a projection of what the DataManager Singleton is observing every second. Below is a system diagram of how this works.

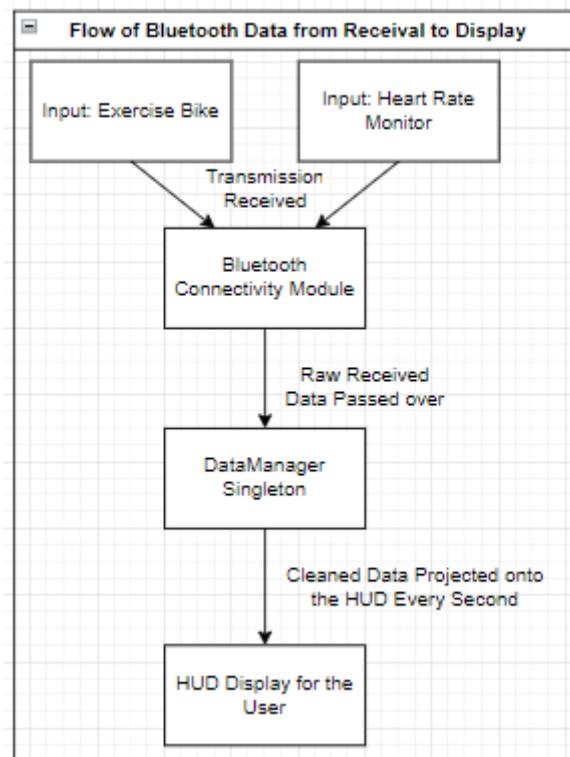


Figure 10 - Diagram Representing Flow of Bluetooth Data to Heads-Up Display

The HUD's logic is simple as it is mainly manipulating TextMeshPro assets within the scene. It comprises of a public method UpdateText() utilised by the DataManager that updates what the text is displaying. An Update() function to be called every frame ensuring that the transform component of the HUD is consistently facing the camera (VR headset position) to ensure that wherever the player is looking the HUD will always be readable. And two public methods that are called when the user presses the two buttons built into the HUD, one of

which is used to toggle the information if the user prefers to not see how they are performing, and the other is used to end the workout and return to the main menu area.

```
10 references
public void UpdateText(TextMeshProUGUI txt, string val)
{
    txt.text = val;
}

0 references
private void Update()
{
    transform.LookAt(cameraToLookAt.transform);
}

0 references
public void ToggleHideData()
{
    if (dataParent == null)
    {
        Debug.LogWarning("Data parent not assigned in editor");
        return;
    }
    dataParent.SetActive(!dataParent.activeSelf);
}

0 references
public void ReturnToTutorialArea()
{
    SessionManager.Instance.SetUndecidedMode();
}
```

Figure 11 - HUD Functionality Code

The DataManager utilises the HUD in the UpdateHUD() method within its class, this method is called once per frame and comprises of updating text for all of the variables that are to be presented to the user. In code form this is a batch of if statements checking if the reference to the TextMeshPro asset in the scene isn't null and then proceeding to call the UpdateText() method if the condition is met.

Once running, the code produces something that looks like the image depicted below, where data is presented to users every frame, and every time the Bluetooth receiver gets new data to be processed, it is presented to the user on the very frame that happens.

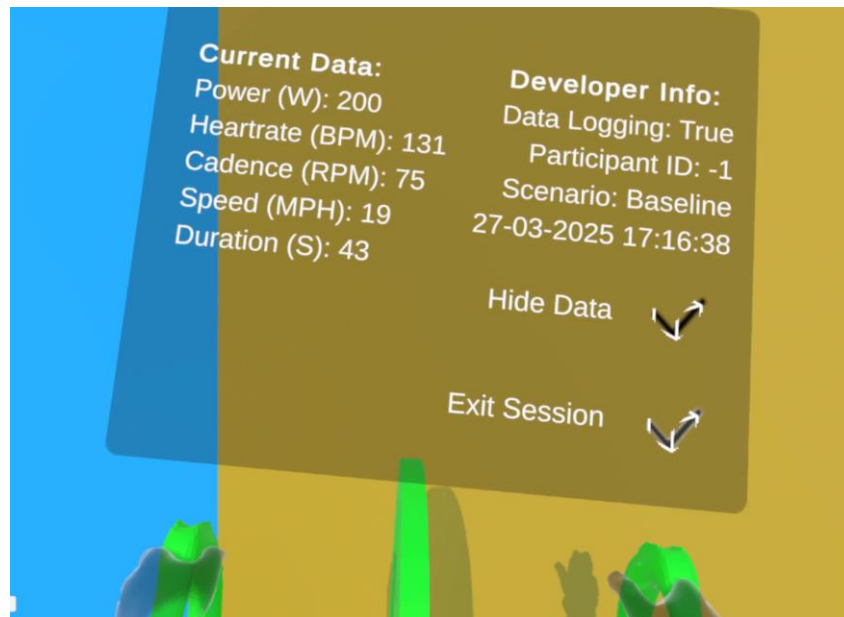


Figure 12 - Final HUD Prototype

3.3.2. Singleton Managers

Data Manager

The DataManager class in this project is a comprehensive system for managing the cycling performance of the user.

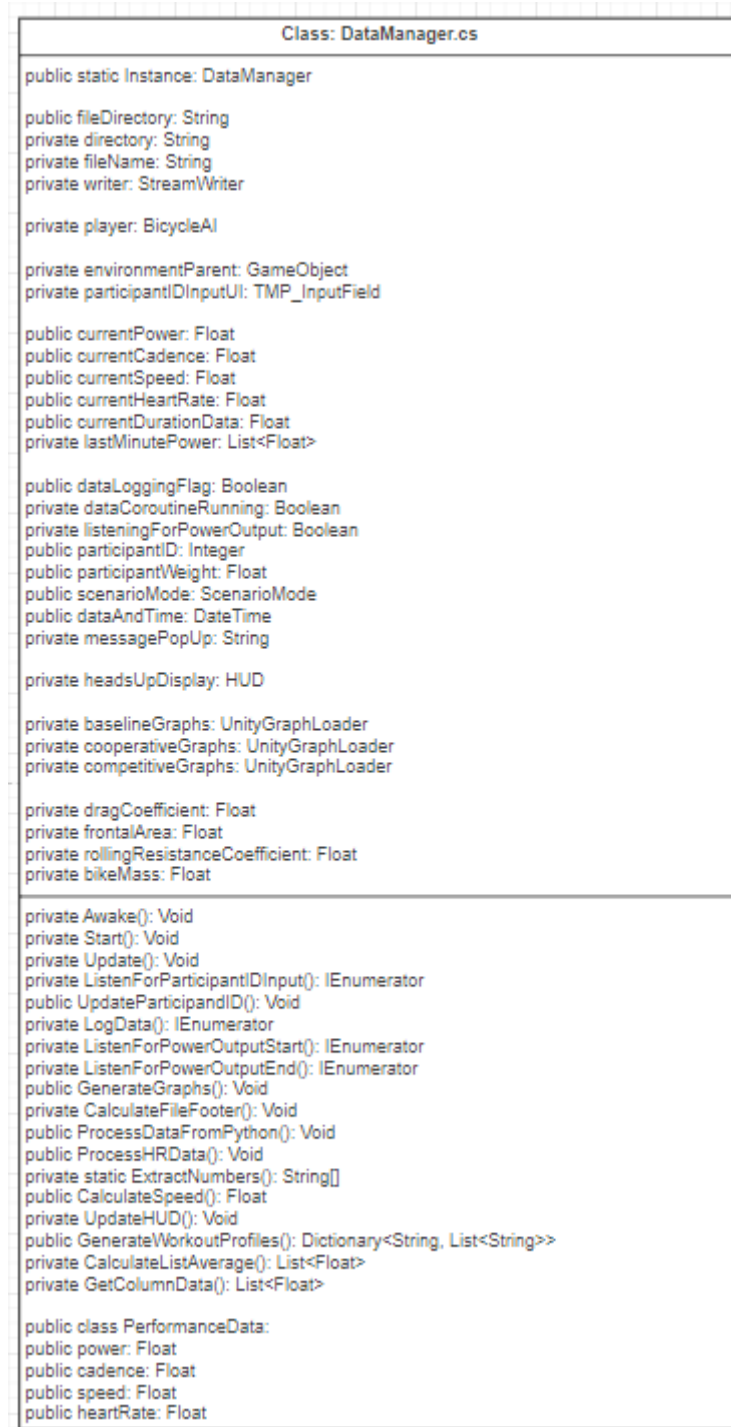


Figure 13 - DataManager Class Diagram

Core Management Functions

The DataManager uses a singleton pattern implementation to ensure that only one DataManager exists and sets itself to DontDestroyOnLoad() to ensure it persists throughout different scenes.

It is responsible for file system management where it creates and manages a directory structure for performance logging. It handles file creation, writing, and parsing data for analysis. It uses a StreamWriter for datalogging into a CSV and exports said CSV after a session.

It monitors power output to automatically start and stop logging data when the user starts and stops pedalling. This is done by coroutines for background data monitoring, the coroutines `ListenForPowerOutputStart()`, and `ListenForPowerOutputEnd()` track active and inactive periods to automatically start and stop sessions based on the users input via the bike.

Data Collection

The `DataManager` records real-time cycling metrics (Power (Watts), Cadence (RPM), Speed (MPH), Heart rate (BPM), Session Duration (Seconds)). It uses physics-based calculations to calculate the speed the bicycle is moving based on power output using physics formulas that account for drag coefficient, frontal area, rolling resistance factors, and cyclist and bike mass. As well as integrates external data from the two Bluetooth sources using the `ProcessDataFromPython()`, and `ProcessHRData()` methods and uses the `ExtractNumbers()` method to use regular expressions to parse numerical data from the strings.

Analytics and Performance Assessment

The `DataManager` calculates performance metrics after the user completes their workout. This is done in the `CalculateFileFooter()` method which takes the spreadsheet and reads the data from it. It produces metrics such as Functional Threshold Power (FTP) estimation, Heart Rate Recovery (HRR) analysis, Efficiency Factor (EF) calculation, Peak Power Output (PPO), and Total Work Done.

It performs a statistical analysis tracking rolling averages like last minute power, as well as calculates session average for all metrics, determines maximum values for performance indicators, and computes power-to-weight ratio when participant weight is available.

It is also responsible for historical data processing, it creates performance profiles from historical data using helper functions that extract specific metrics from the CSV files, as well as processes multiple data sets for averages.

User Interface and Feedback

Heads-Up Display (HUD) integration is controlled through the `UpdateHUD()` method that updates real-time performance data on screen, displays session status information shows participant and scenario information, and presents pop-up messages for system status.

It handles participant ID input via Unity UI features, as well as listens for the input to manage environment activation, and uses the method `UpdateParticipantID()` to validate and process user input.

Data Visualisation

Graph generation is handled through using the external python script `DataVis.py` mentioned earlier and creates separate graphs for different scenario modes

(Baseline, Cooperative, Competitive). Data exportation and reporting is handled using the `CalculateFileFooter()` method adding summary statistics to data files, it creates formatted reports with average and performance metrics and organises the data for external analysis.

Study Management

The `DataManager` handles different scenario modes, manages participant identification, tracks experimental conditions and parameters, and records data and time information for data validity. It also filters out invalid or problematic data entries, removes trailing empty data points, and provides error logging and debugging information.

This singleton serves as a central nervous system for the performance tracking system handling everything from raw data collection to sophisticated analysis and visualisation.

Session Manager

The `SessionManager` is a controller class in the project which orchestrates multiple interconnected systems to create a smooth cycling experience. It serves as the central hub that coordinates scenarios, AI behaviour, player positioning, and session state management.



Figure 14 - SessionManager Class Diagram

Core Architecture

The SessionManager implements a singleton pattern to ensure that only one instance of the class persists throughout runtime. This provides global access through the application via the static “Instance” property. It supports multiple cycling modes through an enum structure called ScenarioMode for each implemented scenario (baseline,cooperative,competitive) as well as tracks scenario state transitions to trigger the appropriate scene setup using Unity prefabs.

XR Integration and Positioning

The SessionManager controls the XR Origin position and rotation for different scenarios, implementing smooth transitions between positions using coroutine-based lerping, and provides XR recenter functionality to realign the virtual experience with physical setup. It also allows toggleability of the UI elements in the VR environment if participants want their live data hidden.

AI Cyclist Management

The SessionManager identifies and filters through AI based on scenario mode and AI roles, organises cyclists into appropriate formations based on the selected scenario, and maintains activeAI list as a primary reference for manageable AI entities. It also assigns different AI

states based on position and scenario, coordinating positional relationships between AI cyclists, and updates AI behaviour parameters in real-time.

Cooperative Mode

The SessionManager implements realistic paceline cycling formations with leader rotation, maintains proper spacing and drafting positions between cyclists, and monitors paceline speed for consistent group movement. The way the rotation logic works is it moves lead cyclists to the back at timed intervals, updates AI states when positions change in the formation, and triggers specialised movement patterns during transitions, like the PeelOffPaceLine() method.

Competitive Mode

The SessionManager uses calculated performance profiles to create realistic racing scenarios, applying individual performance variations set in the Unity inspector to create distinct AI competitors, and adjusts AI performance based on real-time user performance metrics. It modifies AI performance based on the user's current heart rate compared to the heart rate in the historical data on that second of the exercise. It then scales difficulty using percentage based adjustments on the difference between those values to maintain a competitive balance, and falls back to relative-to-player pacing when the recorded data runs out.

Technical Features

The SessionManager uses coroutines for time-dependent processes like paceline rotation, and racing progression, as well as injects second-by-second data to the AI's in the competitive scenario. It also works with Unity's NavMesh system for navigation around the virtual velodrome.

3.3.3. Bicycle Objects (Player and AI)

BicycleAI.cs

BicycleAI.cs serves as a controller for the bicycle gameObject that uses Unity's NavMesh system to navigate around the virtual velodrome. The script manages different bicycle behaviours depending on their role (player,competitor,teammate) and the current game mode (baseline,cooperative,competitive). Its core components include:

AI Types and States

The script manages its behaviour based on the assigned AIType and AIState that can be assigned in the inspector and changed at runtime. There are three AITypes (Player,Competitor,Teammate) and two AIStates (Pulling,Drafting).

Navigation System

The script uses Unity's NavMesh system for pathfinding, and implements a custom path smoothing method using Catmull-Rom splines. It adjusts the speed of the NavMeshAgent attached to the bike based on scenario and positioning of the AI.

Scenario Modes

Depending on the scenario mode currently active, the AI will be assigned different targets to path find to. If the scenario is baseline or competitive the AI is given a target in the virtual environment that is updates every quarter of the track to ensure that the bicycle keeps doing laps around the track. If the scenario is cooperative the AI alternates its target depending on its paceline position, if it is pulling the target is in the virtual environment, if it is drafting the target is the bicycle in front in the paceline.

BicycleSmoothRotation.cs

BicycleSmoothRotation.cs works alongside the BicycleAI.cs script on the bicycle object and is implemented to solve the problem of snap turning with Unity's NavMeshAgent navigation. Without this script the navigation rotation is based on the angular velocity of the object, which if set to a low value works smoothly at low speeds but at high speeds will not be able to follow the path found for the agent and will send it flying off of the edge of the velodrome

as a result. If the angular velocity is set too high, then the agent can follow the path but will jolt every time the agent hits a vertex in the path.

BicycleSmoothRotation.cs handles this problem by adjusting the rotation of the agent itself. It does this by looking ahead on the path and anticipating corners (vertices) and then smoothing the rotation using a quaternion interpolation to create a gradual turn rather than a snap turn, adjusting the turning speed based on the velocity of the bicycle object.

OpacityController.cs

OpacityController.cs is a script implemented to reduce opacity or hide the mesh of the adaptive AI depending on how close they are to the player.

3.3.4. Bluetooth Connectivity

Bluetooth was by far the most tedious part of the project as it went through the most iterations of all the components in the project. Initially, Bluetooth was managed by an FTMS indoor bike script which was present in the Unity scene and utilised the standard BLE library for windows through BleWinrtDLL.dll in the project, this solution was quick and easy to set up with really only the device UUID's and main execution sequence to customise to the specific use case for this project. This worked great for receiving the bike data and receiving the heartrate data once all the correct values were plugged in. However, it did not work so great when you try to do them both asynchronously. The methods implemented in the BleApi have a parameter that can be passed through which enables or disables a block to prevent other devices from using the BLE service, despite these parameters being set to false they still would not both run at the same time. To try to combat this two separate scripts were written for bike and heart rate data and placed on separate objects in the Unity scene, this did not solve the problem. The next attempt was to utilise threading to prevent the BLE processes from interfering with one another, however this also did not work.

In the end the solution in the final prototype uses two separate Bluetooth libraries on two separate programs and two separate receival mechanisms. The heartrate data is received via the BleApi in the BleWinrtDLL.dll file, and the exercise bike data is received by a separate python script which then passes the received data into Unity via websockets. The data is then passed to the DataManager for cleaning and processing. This solution was not ideal as it was desired that the Bluetooth functionality remained entirely within the unity project to prevent unnecessary complexity, but due to time constraints this was the solution that was stuck to.

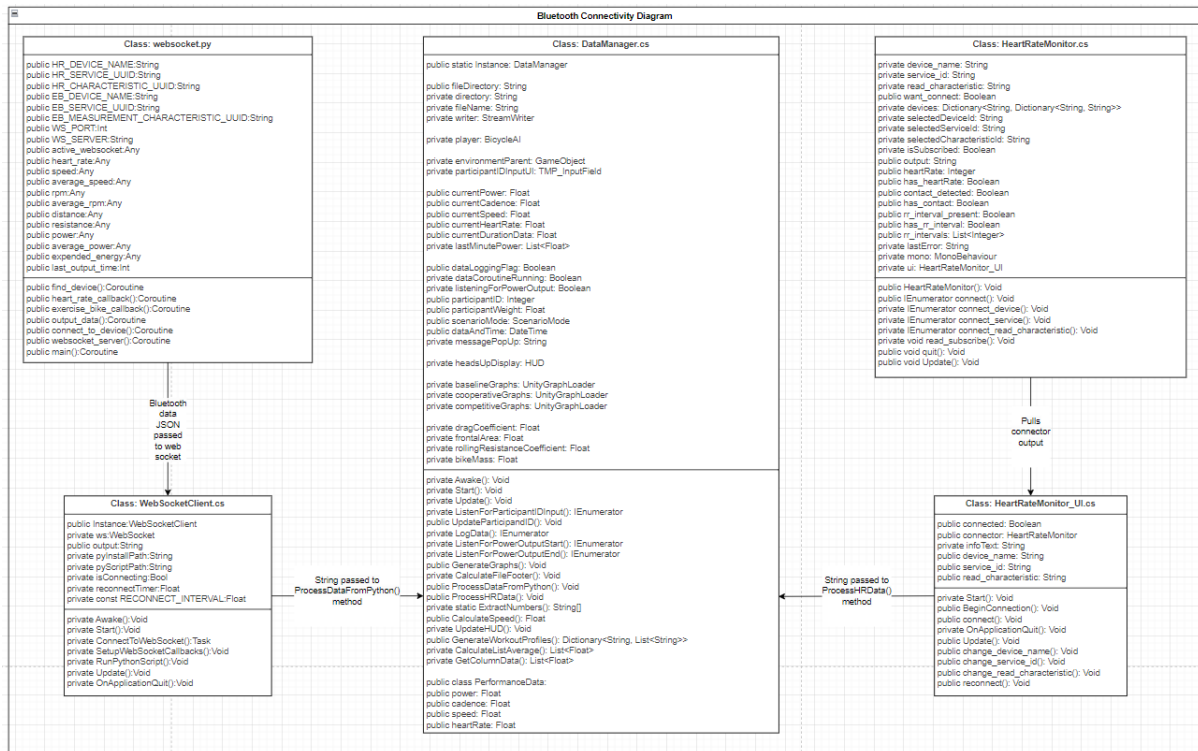


Figure 15 - Bluetooth Connectivity Diagram

Exercise Bike Connectivity

```

async def main():
    """Runs the BLE and WebSocket servers concurrently."""
    server = await websockets.serve(websocket_server, "0.0.0.0", WS_PORT)
    print(f"WebSocket Server running on {WS_SERVER}")

    await asyncio.gather(
        connect_to_device(EB_DEVICE_NAME, EB_SERVICE_UUID, EB_MEASUREMENT_CHARACTERISTIC_UUID, exercise_bike_callback),
        server.wait_closed()
    )

if __name__ == "__main__":
    loop = asyncio.new_event_loop()
    asyncio.set_event_loop(loop)
    try:
        loop.run_until_complete(main())
    except KeyboardInterrupt:
        print("Script terminated by user.")
    finally:
        for task in asyncio.all_tasks(loop):
            task.cancel()
        loop.run_until_complete(asyncio.gather(*asyncio.all_tasks(loop), return_exceptions=True))
        loop.close()

```

Figure 16 - websocket.py main execution snippet

Above is a snippet of the main execution within the websocket.py file. When the program runs you can see that an asynchronous loop is set on the main function to run continuously until a keyboard interrupt or closure of the terminal by the user. What this does is set up the websocket server and starts connecting to the exercise bike, making use of `asyncio.gather()` to run them both concurrently.

This JSON message is then received by the WebsocketClient.cs script in the Unity project. The purpose of WebsocketClient is to connect to the WebSocketServer and to parse and process the FTMS data received from the python script.

Heart Rate Connectivity

Heart rate connectivity is handled by two scripts written in C#. the first one is the connector (HeartRateMonitor.cs) which utilises the BleApi within the BleWinrtDLL.dll library. This class encapsulates the full BLE connection and data reading process for heart rate monitors in Unity. It manages connection flow, data interpretation, error handling, and UI updates.

The second class to the heart rate connectivity implementation is the HeartRateMonitor_UI class that acts as a UI controller for the HeartRateMonitor class. It manages the heart rate monitor connection, updates heart rate data each frame, and provides interface methods to update connection parameters via Unity's UI. It is to be attached to a GameObject in a Unity scene. Below are the methods implemented in this class and their functionality:

To summarise, the HeartRateMonitor_UI class is the script that bridges the BLE logic handled by the HeartRateMonitor script with the Unity editor and the game logic that consumes the HR data. This class ensures the app automatically connects on launch and keeps BLE processing in sync with the Unity game loop.

3.3.5. Analysis.py

Analysis.py is a script implemented for the sole purpose of processing the data collected from the user study. Including plotting moving averages (Power, Heart rate) for each participant across all three scenarios, generating boxplots for both the RPE scale and IMI inventory (both overall, and subscale), as well as calculating mean scores from the post-scenario questionnaire. It contains loaders for each individual spreadsheet to ensure that the data is correctly formatted into Pandas Dataframes, as well as a main execution that runs functions to generate each plot.

3.4. User Study

SREC reference: COMSC/Ethics/2025/010

Date approved: 26/03/2025

Participant Demographics: The study conducted comprised of 6 Males, 6 Females between the ages of 18-35.

Inclusion Criteria:

- Age: 18 years or older.
- Must be in generally good health, with no known medical conditions that would make high intensity cycling unsafe.
- No history of severe motion sickness or epilepsy triggered by VR.
- Open to all fitness levels, from beginners to advanced cyclists.
- Comfortable using a VR headset, but no prior VR experience is required.
- Must be able to complete three cycling scenarios within the study timeframe.

Exclusion Criteria:

- Individuals with heart disease, uncontrolled high blood pressure, respiratory issues, or joint injuries that could be aggravated by cycling.
- If prone to nausea, dizziness, or disorientation in virtual environments.
- Epilepsy or other conditions that could be triggered by flashing visuals in VR.
- Pregnancy as a precautionary measure for participant safety.
- Participants who cannot wear a VR headset comfortably.
- Anyone who does not agree to the informed consent process.

Prior to participation participants read the Participant Information Sheet and signed Informed Consent Forms. Sessions comprised of participants completing one of the three scenarios whilst their performance data was collected in real-time (Power, Cadence, Heart Rate). After completing the exercise participants filled out Microsoft Forms created for each of the surveys within the ethics application:

- The demographic questionnaire asked participants their age, sex, and gender, as well as their study ID so group data anonymously and was completed after the first visit.

- The Borg RPE Scale asked participants to rate their perceived level of exertion and was completed after every visit.
- The Intrinsic Motivation Inventory asked participants 1-7 scale questions on choice, competence, effort, enjoyment, fun, interest, pressure, and value and was completed after every visit.
- The VR Cycling Ramp Test Post-Scenario Questionnaire asked participants likert scale questions on their enjoyment, if they felt motion sick etc, and collected qualitative feedback on what participants thought what was good, what was missing, what made them push harder etc. This questionnaire was completed after every visit.
- The Big Five Inventory asked participants questions on the Big Five Personality Test and was used to look for correlations of particular personality types favouring and outperforming in particular social dynamics. This was completed after the final visit.

Participants all completed the baseline scenario in their first visit, and then to mitigate order effect half of the participant pool completed the cooperative scenario second, and the other half completed competitive.

3.4.1. Risk analysis and mitigation strategies

Physical Risks and Management

Participants were advised not to take part if they had any medical conditions that could be worsened by intense exercise (e.g. heart conditions, uncontrolled asthma).

A full briefing was provided before starting the ramp test, explaining the nature of the test and that they should stop immediately if they feel unwell, faint, or in pain beyond muscular fatigue.

The researcher monitored participants throughout the session, watching for signs of distress and would stop the test if necessary.

Psychological/Emotional Risks and Management

Participants were informed that while the task is designed to be challenging, they were under no obligation to complete the test to exhaustion and could stop at any time without penalty. The competitive/team dynamics were explained carefully to ensure participants know that their performance wasn't being judged and that their effort was what matters, not their final score.

After the test, participants were debriefed to ensure that any negative feelings were addressed, and support resources were provided if required.

VR-Specific Risks and Management

Before participation, participants were informed about the possibility of motion sickness when using VR.

Participants were advised to stop immediately if they begin to feel dizzy, nauseous, or disoriented.

The software was tested to ensure minimal motion lag and smooth visuals, and short VR exposure times were used to reduce risk.

Breaks were offered as needed.

No form of equipment malfunction could cause harm to the participant, in the event that data collection devices fail at runtime the session would conclude, and the participant would be requested to return at a different time when they are well rested.

3.4.2. Timeline for development testing and study

Development began at the beginning of February 2025 and concluded in the middle of April 2025. Participants were then recruited and participated all throughout April with a self-imposed deadline of 30/04/2025. Unfortunately, due to many of participants being students, and easter break starting on 12/04/2025, some participants were not able to finish all three visits in time. Only participants that have completed all three visits will be used in the

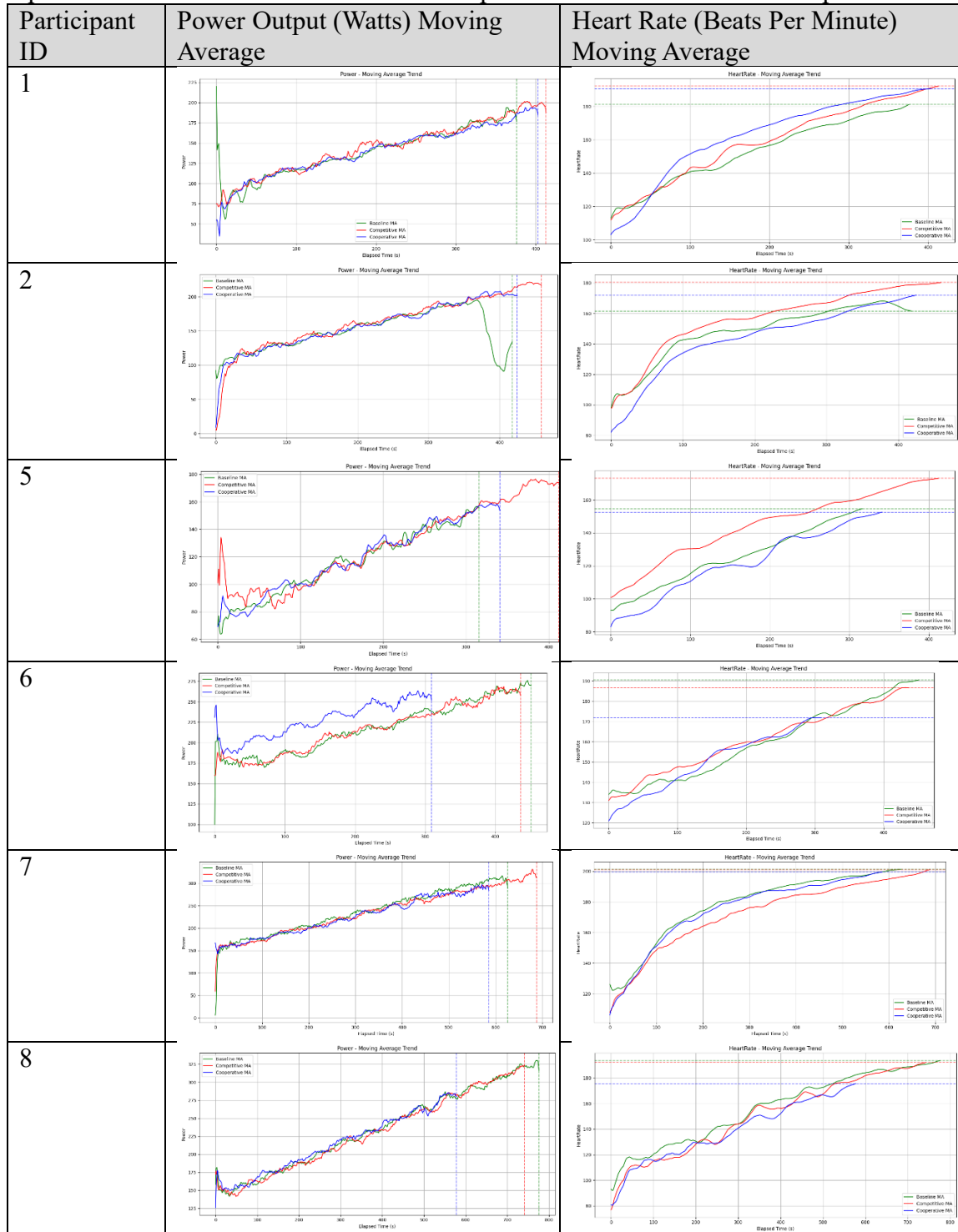
comparison data (Performance, RPE, IMI, BFI) but all qualitative data from participants will be used when evaluating the project (Post-Scenario Questionnaires).

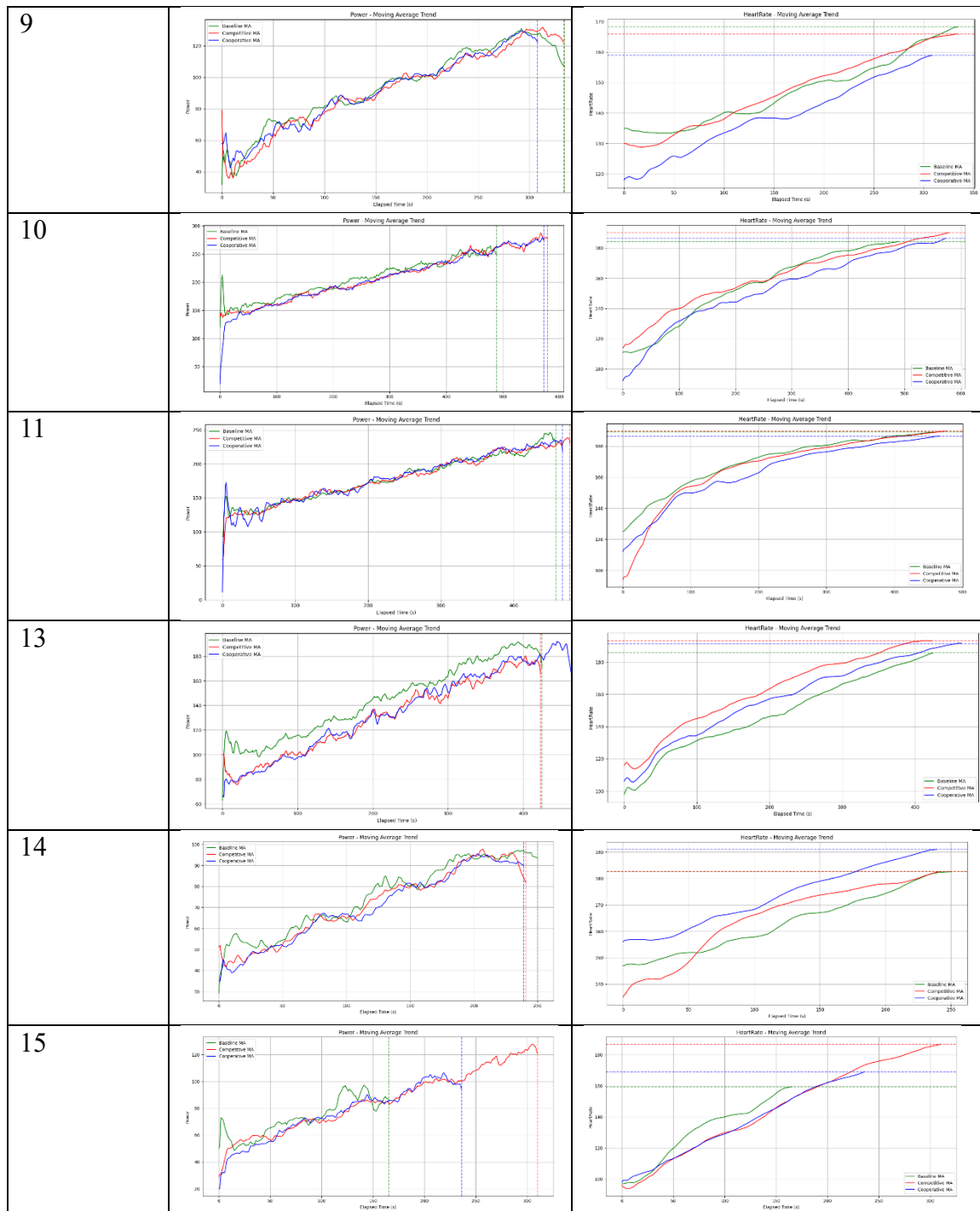
4. Results and Evaluation

4.1. Quantitative results

4.1.1. Moving Averages Across the Three Scenarios

Moving average graphs were calculated using the real-time performance data saved to .csv format after each session. Below are the plotted graphs from the analysis.py script colour coded by scenario (Green = Baseline, Blue = Cooperative, Red = Competitive). Any missing participant numbers were users who did not complete all three scenarios for comparison.





4.1.2. Interpretation of results regarding performance under cooperative/competitive dynamics.

Generally, graphs between baseline, cooperative, and competitive show that when presented with social dynamic participants are more willing to further exert themselves over the same workout. The data suggests that the use of cooperation in VR exergaming does not necessarily present greater performance compared to solo cycling, however the use of competition does with only one participant performing better in cooperative than competitive.

4.1.3. Rate of Perceived Exertion Scale Findings

The Rate of Perceived Exertion Scale asked participants to rate the level they felt they exerted themselves in each scenario.

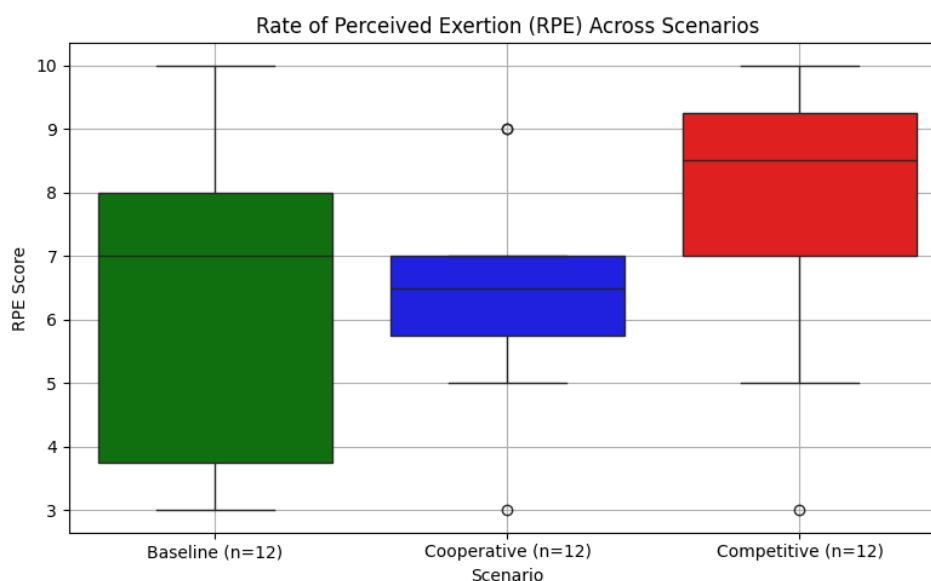


Figure 17 - Rate of Perceived Exertion Box Plot

As you can see from the box plot above, participants felt that they clearly exerted themselves harder in competitive compared to solo or cooperative cycling, with the participants generally reporting lower perceived exertion in cooperative compared to baseline.

4.1.4. Intrinsic Motivation Inventory Findings

The Intrinsic Motivation Inventory asked participants to rate how much they agreed with statements on a Likert scale (1-7), these statements can be classified into psychological dimensions related to exercise motivation and experience. Below are said statements and their classifications:

- 'I enjoyed engaging in my session very much.': 'Enjoyment1',
- 'After my session, I feel motivated to reach my goals.': 'Motivation',
- 'I put a lot of effort into my session.': 'Effort1',
- 'Engaging in my session was enjoyable.': 'Enjoyment2',
- 'I believe that the method I used to track my workout could be of value to my exercise goals.': 'Value',
- 'I believe I am pretty good at working out.': 'Competence',
- 'I tried very hard during my exercises.': 'Effort2',
- 'I found my session to be interesting.': 'Interest',
- 'I engaged in my workouts because I wanted to.': 'Choice',
- 'I had fun during my session.': 'Fun',
- 'I felt very tense during my session.': 'Pressure1',
- 'I experienced a lot of pressure during my session.': 'Pressure2'

These were then grouped into subscales for visualisation via radar chart.

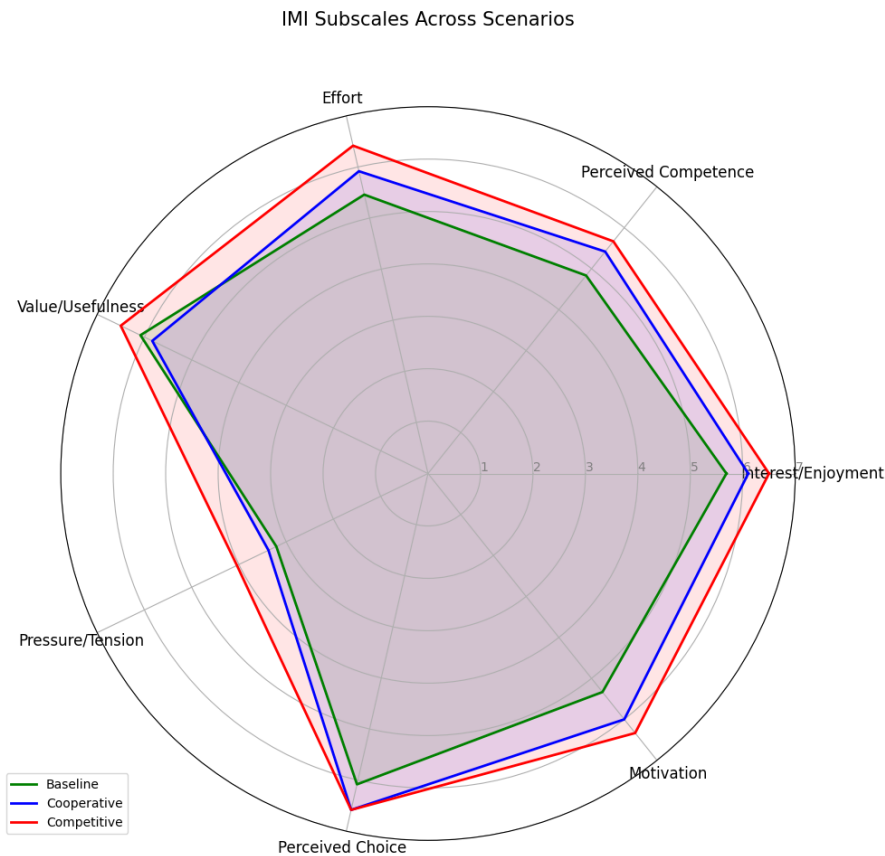


Figure 18 - Intrinsic Motivation Inventory Radar Chart

As you can see from the radar chart above, after both sessions with social dynamics participants said they felt more motivated in virtually all subscales compared to the baseline scenario, with the competitive scenario surpassing the cooperative in virtually all dimensions suggesting that on average competition has a greater impact on motivation than cooperation compared to solo cycling.

4.2. Qualitative Results

4.2.1. Post-Scenario Questionnaire

The post-scenario questionnaire comprised of a mix of Likert scale and open-ended questions. Below is a walk-through of the findings for each question.

Likert scale questions:

1.	I felt motivated to push myself harder because of the VR scenario I was in.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
4.4	5.17	6.33	
2.	The way the scenario was designed encouraged me to give more effort than I normally would with a standard exercise bike workout.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
4.8	5.42	6.5	
3.	I pushed myself harder than I usually would when cycling on my own.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
5.53	5.75	6.67	
4.	The VR environment helped distract me from the discomfort of the workout and encouraged me to keep going.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
4.67	5.17	5.92	
5.	If you were in the cooperative scenario: The presence of AI teammates made me want to contribute more and work harder for the team effort.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
N/A	5.17	N/A	
6.	If you were in the competitive scenario: Racing against AI opponents made me push harder to avoid "losing" to them.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
N/A	N/A	6.5	
7.	If you were in the baseline scenario: I felt less motivated to push myself because I was cycling alone.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
4	N/A	N/A	
8.	Compared to a standard gym workout; this VR scenario made me work harder.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
5.27	5.75	6.58	
9.	The VR experience made the workout feel more purposeful than traditional workouts.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
4.93	6.08	6.58	
10.	The VR system was easy to understand and use.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
6.13	6.33	6.33	
11.	The VR system felt intuitive and did not require much effort to figure out.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
5.87	6	6	
12.	I felt fully immersed in the VR environment during the ramp test.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
5.33	5.92	6.33	
13.	The graphics and visual elements were clear and helped me understand my progress or position in the scenario.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
5.4	5.92	6.25	
14.	I experienced feelings of dizziness, nausea, or motion sickness while using the VR system.		
	Baseline Mean Score	Cooperative Mean Score	Competitive Mean Score
3.6	2.58	2.17	

Figure 19 - Post Scenario Questionnaire Likert Results

As you can see from the mean score across the scenarios, the competitive dynamic received higher Likert scores in virtually all questions besides experiencing motion sickness.

Open ended questions:

What aspects of the VR scenario made you push harder (if any)?

It became abundantly clear from responses to this question that there were two main driving motivators within the program that made participants want to push harder:

- **The implementation of the Heads-Up Display (HUD)**, participants gave responses such as “the stats on the screen helped me keep track and work harder”, “Seeing my stats on the screen motivated me to push harder”, specifically participants knowledge of the workout duration “seeing time to aim for, heart rate, and the track to follow”, “being able to see the timings made me want to work harder”, “beating the time”.
- **The implementation of interactive feedforward**, participants gave responses such as “wanting to catch up to the next cyclist”, “Some other AI cycling with me and I want to beat them”, “Having some AI riding in front of me, even I coop mode I feel competitive”, “Knowing that a slightly improved (2-5% ?) version of me was ahead made me want to catch up to be the slightly fitter version of myself”.

What could be improved in this VR scenario to make you feel even more motivated to give your maximum effort?

The participant feedback for this question can be organised into five categories:

- **Enhanced Environmental Design:** several responses said they would feel more motivated if there were changes to the environment of the application, “A more detailed surrounding”, “more immersive vr environment”, “More engaging map like zwift for extra stimulation/distraction from discomfort/exertion”, “If the scenario provide different sceneries instead of looping in the same scene, it will be more enjoyable”.
- **Improved Cooperative & Competitive Elements:** some participants wanted better competitive mechanics, like leaderboards and distance counters. Multiple responses focused on improving AI behaviour and positioning, " There's lag when the AIs are getting back to leading the player, would be more responsive when the is unnoticeable? It feels disconnected every time the lag happens”, “maybe more pacers, i felt like i was never able to catch the ai ahead of me. it could've been made more realistic if the ai went around me to overtake rather than through me. this could have made me see them coming a bit better and make me push as they try to overtake”. From an observational perspective as the developer, it was noticeable in the study that the consistency of the AI with regards to the distance from the player in the competitive scenario was heavily down to the users ability to maintain a consistent cadence and power output.
- **Improved Gamification:** multiple participants suggested the implementation of visible progress bars, collectable items, and virtual rewards or prizes for effort. “a way of goal setting to see progress visually ie progress bars”, “buttons to collect like mario kart”, “vr prizes for how hard you push”.
- **User Experience Improvements:** music and bike customisation was suggested as a motivational element, as well as techniques to reduce motion sickness. “If I can choose different bikes, scenario and AI. For example, use my friend data as AI and ride in a forest environment.”, “reducing of motion sickness”.
- **Motivational Features:** participants suggested features like notifications for decreased performance to encourage consistent effort, better placement of AI competitors to ensure targets being attainable, as well as less repeatability in cooperative scenarios. “could include motivation/notifications for decreased cadence to encourage the cyclist to keep up the pace”, “maybe having a few cyclist closer to feel like you're achieving something but also some far ahead for motivation”, “i would have liked if there was a way to earn your way to be at the front of the group - I

figured that it just changes spaces every 10 seconds which can become a bit predictable.”.

Any other comments about your experience?

Responses to this question generally followed the same narrative of the previous one, however with greater mention of real user competition rather than AI.

If you experienced any discomfort (e.g. motion sickness, dizziness), please describe what you felt and when it occurred.

Responses to this question clearly highlighted the patterns in the design of the program that caused motion sickness. A significant majority of the responses that experienced discomfort put it down to the velodrome corners as the primary cause. This suggests a vestibular mismatch between the visual input and the physical sensation of the user during turning in the virtual environment, “a little bit dizzy when I was more tired and going around corners”, “slightly dizzy when going round corners”, “felt seasick, specifically on the corners of the velodromes”. Additionally, several participants reported VR related discomfort as the workout intensity increased, “motion sickness when the workout got harder”, “As it got harder the motion sickness got worse”, “a little bit dizzy when I was more tired and going around corners”. Notably a few participants said that discomfort was reduced after the first session suggesting the social elements may have provided a distraction to the discomfort/motion sickness, “barely felt any motion sickness compared to last time i did it.”, “I didn't feel at all motion sick on this one, whereas I did a bit on the baseline - maybe because I was distracted by the people in front”.

4.2.2. Big Five Inventory Findings (if any)

No correlations between personalities could be identified from the used participant pool as virtually all lines of motivational and engagement questioning pointed that competitive social dynamics are more favourable than cooperative, but both are more favourable than solo cycling. However, there were anomalies where some users performed better in their first session than the following two but due to the small study sample size this would need to be further explored before drawing conclusions.

4.3. Evaluation and critical reflection

4.3.1. Effectiveness of gamification and AI in enhancing engagement

The RPE boxplot and IMI radar chart results show concrete evidence that the use of both AI teammates and competitors makes the player feel more motivated in virtually all subscales of motivation, as well as more willing to exert themselves and push themselves further. Whilst the implementation of a Heads-Up Display (HUD) serves as a strong motivational foundation for a user to push themselves, feedback from the post-scenario questionnaire suggests that more gamification elements could prove to help increase motivation and engagement.

4.3.2. Effectiveness of motion sickness mitigation

Motion sickness was indeed an issue encountered during the study, a few of the participants said they felt motion sick throughout the workout, expressing that it would worsen as the effort went on. However, a many of the participants who completed multiple visits said that they felt the effect less when in AI scenarios. This suggests that either it is quick to overcome the initial motion sickness of the solution, and/or that the use of AI as a focus point for the user creates a distraction from the effect.

4.3.3. Success and limitations of AI difficulty adjustment

The cooperative scenario AI performed as intended in the scenario for all participants, irrelevant of participant power output the AI moved around the user in a structured formation and kept a tight group. Some participants said that the rotation mechanism was displeasing to the eye and felt that it was distracting them from getting into a level of exercise flow state, this should be further refined for future work. Overall, the data from the post-scenario

questionnaire and the performance plots suggests that the use of AI NPC's in the cooperative scenario did increase users want to push harder for the “team effort”.

The effectiveness of the interactive feedforward in the competitive scenario was inconsistent for participants. Some participants found that they were able to catch their competitors and said that it was a primary driver for their effort. Others said they thought that the competition should have been closer to them to really get that effect. Overall, the mechanism did what it was meant to but the effectiveness of it should be further refined to consider for a variety of user input styles. The responses to the post-scenario questionnaire show that the presence of the competition strongly made participants want to push harder to avoid “losing” to them.

4.3.4. Discussion of accessibility and usability feedback from participants

When presented with a question asking participants if they felt the VR system was easy to understand and they understood what they were meant to do, across all scenarios the feedback averaged a score above 6 on the Likert scale. This shows that participants felt that the user interface was intuitive and explained the functionality and what was asked of them well.

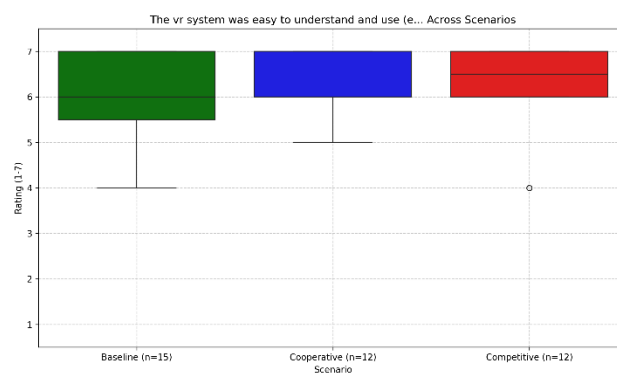


Figure 20 - System Usability Likert Results

When presented with a question asking about the intuitiveness and effort required to navigate the program users all on average agreed that the system did not require much effort to figure out. Additionally, participants agreed on average that the graphics and visual elements of the program were clear and helped them understand their progress and position in the scenario.

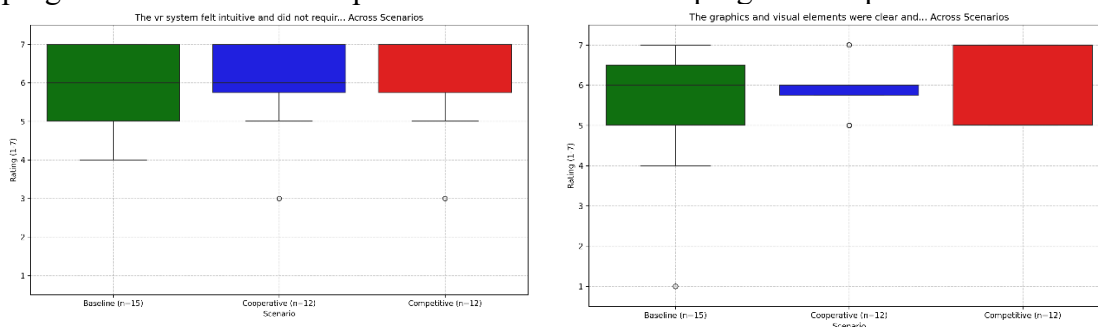


Figure 21,22 - System Clarity & Intuitiveness Likert Results

4.3.5. Actionable recommendations for exergame design

Designing for User Engagement and Motivation – It is recommendable to incorporate rewards systems and progression systems to keep the user engaged and motivated to keep exercising. Traditional points-based systems could suffice for these allowing users to collect units of a game currency to keep them hooked. Diversity and expand users’ surroundings/environment to pose as a distraction from the workout and to potentially reduce motion sickness.

Designing for Accessibility and Inclusivity – Offer adjustable settings for physical abilities, adaptive AI is a good idea however requires refinement to make it perfect for the user. Ensure a simplistic and intuitive user interface that doesn't require preexisting VR knowledge to operate. Implement controller-free solutions for ease-of-use as controller experiences are impractical with exercise.

Designing for Data Tracking and Real-time Feedback – Utilise Heads-Up Display's with toggleable visibility to keep users incentivised with their own performance. Utilise logging methods outside of writing in real-time to spreadsheets as over long workouts it can become damaging to performance, use JSON storage methods instead.

Designing for Gameplay Features – Implement multiplayer connectivity to boost user motivation. Use long grace periods at beginning of workouts involving historical AI to cancel out any differences in warmups compared to training data. Straight line maps are more favoured for accessibility as inexperienced VR users find corners to be a cause of motion sickness.

4.3.6. Primary Issues encountered

Bluetooth connectivity – The BLE.dll's inability to manage two connections asynchronously caused overcomplication in the Bluetooth component of this project. The current use of BLE libraries would be impractical in a full solution, it would be better if an adaptable plugin was defined for use before implementation.

AI Questionable Reliability – The competitive AI mechanics were hit and miss and relied too heavily on the user's heartrate for modification, if the user began the workout at even a slightly lower heartrate it would give the AI an overly significant head start hindering the user's ability to catch them.

Data Logging Performance Concerns – For particularly active participants who could last longer in the sessions, frame rate drops were experienced later into the session. For future use algorithmic time complexity should be optimised to prevent such performance issues.

5. Conclusions and Future Work

5.1. Summary of Key Findings

This project set out to research how cooperative and competitive social dynamics influence user engagement, intrinsic motivation, and performance. The user study found that the implemented social dynamics both increased intrinsic motivation and engagement. No correlations between particular personalities could be identified from the used participant pool as virtually all lines of motivational and engagement questioning pointed that competitive social dynamics are more favourable than cooperative, but both are more favourable than solo cycling.

Additionally, it was found that participants desired increased gamification and namedropped traditional concepts like points-based systems to be more of a driver of exertion for them. Participants also enjoyed the use of a Heads-Up Display as it gave them clear knowledge of their performance and the timer feature was noted as a particularly key motivator.

5.2. Contributions to the field

This project presents a contribution to the field of VR exergaming through the novel integration of dynamic social mechanics. By implementing adaptive AI NPC teammates and competitors within a virtual velodrome environment, this study demonstrates an innovative approach to exercise motivation that doesn't require multiple participants per session. The prototype system's ability to create realistic social dynamics through real-time performance monitoring and adjustable AI behaviour represents a strong foundation for adaptive AI in exergaming design.

Unlike traditional VR exercise applications that rely on predetermined challenges, this research introduces responsive virtual entities that adapt to the user's performance level,

creating more engaging and personalised exercise experiences. The implementation of cooperative formations and competitive pacing mechanism provides a foundational framework that future VR exergame developers can build upon to enhance user engagement.

5.3. Future Work

Improving AI – As stated earlier, the current AI serve as a strong foundation for social dynamic implementation in exergames. However, there is plenty that can still be done to better motivate and engage the user. If the time constraints on this project weren't so limiting, the implementation of a more comprehensive AI system would be used, which would take into account not only individuals performance, but how their cycling style, and strengths and weaknesses (e.g. endurance, interval training, etc).

Improving Gamification & Environmental Elements – Expanding the use of gamification in the project would be highly beneficial as it was requested by participants. Specifically points-based progression systems, achievements, or rewards that could keep users motivated and engaged. Improvements to the environment would likely also improve motivation and engagement, the use of virtual fans/spectators with cheering sound effects and other scene clutter like cars, streetlights, and buildings would be effective in fulfilling the requests from participants.

Multiplayer Functionality – Several participants said they would like to see multiplayer connectivity to allow them to race real people online. This could increase motivation and engagement as some users may find comparison against others more mentally engaging than comparison against themselves. Skills-based matchmaking would be an effective feature for this functionality as it maintains the core requirement of accessibility and adaptability depending on users' performance.

6. Appendices

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